

Mark schemes

Q1.

(a) $\cot x = 2 \Rightarrow \tan x = 0.5$

M1

$x = 0.46, 3.61$

AWRT; no others within range

A1

2

(b) $\operatorname{cosec}^2 x = \frac{3 \cos x + 4}{2}$

$2(1 + \cot^2 x) = 3 \cot x + 4$

Correct use of $\operatorname{cosec}^2 x = 1 + \cot^2 x$

M1

$(2 \cot^2 x - 3 \cot x + 2 - 4 = 0)$

$2 \cot^2 x - 3 \cot x - 2 = 0$

AG; correct with no slips from line with no fractions

A1

2

(c) $(2 \cot x + 1)(\cot x - 2) (= 0)$

Attempt to solve

M1

$\cot x = -\frac{1}{2}, 2$

A1

$\tan x = -2, 0.5$

$x = 0.46, 3.61, 2.03, 5.18$

2 correct Allow 3.6(0)

B1

*4 correct (with no extras in range) AWRT
SC Degrees*

$$\left. \begin{array}{l} 26.57, 206.57 \\ 116.57, 296.57 \end{array} \right\} \text{ B1 For 2 correct}$$

B1

4

[8]

Q2.

(a) (i) $R = 10$

$$R = 10$$

B1

$$\tan \alpha = \frac{8}{6}, \quad \alpha = 53.1$$

For α ; ft incorrect R

B1F

2

(ii) $\sin (2x + 53.1) = 0.7$

M1

$$2x + 53.1 = 44.4$$

one correct answer ; ft α and R

A1F

$$135.6 \text{ or } 135.7, 404.4, 495.6 \text{ or } 495.7$$

3 other correct answers – ignore extras

A1

$$x = 41.2 \text{ or } 41.3, 175.6 \text{ or } 175.7, \\ 221.2 \text{ or } 221.3, 355.6 \text{ or } 355.7$$

four solutions

CAO (with decimal place discrepancies)

Answers only: 0/4

A1

4

(b) (i) $\sin 2x = 2 \sin x \cos x$

B1

$$\cos 2x = \cos^2 x - \sin^2 x$$

identities for $\sin 2x$ and $\cos 2x$ in any correct form

B1

$$\frac{\sin 2x}{1 - \cos 2x} = \frac{2 \sin x \cos x}{1 - (1 - 2 \sin^2 x)} =$$

use of candidate's double angle formulae

M1

$$\frac{2 \sin x \cos x}{2 \sin^2 x} = \frac{\cos x}{\sin x} = \frac{1}{\tan x}$$

AG, CSO

A1

4

(ii) $\frac{1}{\tan x} = \tan x \quad \tan x = \pm 1$
*(see * below)*

M1A1

$x = 45,$

B1

135, 225, 315

if answers given without working, B1 max

*$\frac{1}{\tan x} = \tan x$ seen and followed by
 correct answers without working 4 out of 4*

A1

4

** If hence ignored, so working in sines and cosines, must simplify as far as:*

$\cos^2 x = \sin^2 x \quad \text{or} \quad \cos^2 x = \frac{1}{2} \quad \text{or} \quad \sin^2 x = \frac{1}{2}$					for M1
$\cos 2x = 0 \quad \text{or} \quad \cos x = \pm \frac{1}{\sqrt{2}} \quad \text{or} \quad \sin x = \pm \frac{1}{\sqrt{2}}$					for A1

[14]

Q3.

(a) $2x = 48$

PI by $x = 24^\circ$

B1

$$2x = 180 - 48$$

Accept equivalents for x

M1

$$2x = 360 + 48 \text{ and } 2x = 360 + 180 - 48$$

Accept equivalents for x

M1

$$x = 24^\circ, 66^\circ, 204^\circ, 246^\circ$$

CAO; need all four, no extras in given interval

A1

4

(b) $\frac{\sin \theta}{\cos \theta} = \tan \theta$

Stated or used

M1

$$2\sin \theta - 3 \cos \theta = 0 \Rightarrow \tan \theta = 1.5$$

A1

$$\theta = 56.3^\circ$$

Condone > 1dp

A1

$$\theta = 56.3^\circ + 180^\circ = 236.3^\circ$$

*Ft on c's PV + 180° dep only on the M1
provided no 'extra' solutions in the given
interval.*

A1F

4

[8]

Q4.

(a) $x = \cos^{-1} \frac{1}{3}$

PI

M1

$$= 1.23, 5.05 \text{ (} 0.39\pi, 1.61\pi \text{)}$$

*AWRT (−1 for each error in range)
SC 70.53, 289.47 B1*

A1,A1

3

(b) $\sec^2 x - 1 = 2 \sec x + 2$
use of $\sec^2 x = 1 + \tan^2 x$

M1

$\sec^2 x - 2 \sec x - 3 = 0$
AG; CSO

A1

2

(c) $\sec^2 x - 2 \sec x - 3 = 0$
 $(\sec x - 3)(\sec x + 1) = 0$
attempt to solve

M1

$\cos x = \frac{1}{3}$ or -1 o.e

A1

$x = 1.23, 5.05,$
(2 answers in range from (a)) AWRT

B1F

3.14 (π)
all correct and no extras in range
SC 70.53, 289.47, 180 B1

B1

4

[9]

Q5.

(a) $\{x =\} \cos^{-1} (0.3) = 1.266.... \{= \beta\}$
 $\cos^{-1} (0.3)$ PI by eg 72° or 73°

M1

$\{x =\} 2\pi - \beta$
Condone degrees or mix.

m1

$x = 1.27, 5.02$

Accept 1.26 to 1.27 with 5.01 to 5.02 inclusive

		A1	3
(b)	(i) $m(\pi, -1)$ <i>B1 for each coordinate</i>		
		B1;B1	2
	(ii) $\{x_Q\} = 2\pi - \alpha$ <i>OE (unsimplified)</i>		
		B1	1
(c)	Stretch (I) in x -direction (II) scale <i>Need(I) & one of (II),(III)</i>		
		M1	
	factor $\frac{1}{2}$ (III)		
		A1	2
(d)	$\cos 2x = \cos \frac{4\pi}{5} \Rightarrow 2x = \frac{4\pi}{5}$ <i>OE. (From correct work)</i>		
		B1	
	$\Rightarrow x = \frac{2\pi}{5} (= \alpha)$ <i>Condone decimals/degrees</i>		
	$x = \pi - \alpha$; OE		
	OE eg $2x = 2\pi - \frac{4\pi}{5}$ <i>Correct quadrant; condone degrees/decimals/mix</i>		
		M1	
	$x = \pi + \alpha$; $x = 2\pi - \alpha$; OE <i>Need both (OE for $2x =$) with no extras (quadrants) within the given interval. Condone degrees/decimals/mix</i>		

m1

$$x = \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{7\pi}{5}, \frac{8\pi}{5}$$

*Need all 4 solutions for x but condone unsimplified provided in terms of π
Ignore extra values outside the given interval.*

A1

4

[12]

Q6.

(a) (i) $2 (\operatorname{cosec}^2 x - 1) + 5 \operatorname{cosec} x = 10$

M1

$$2 \operatorname{cosec}^2 x - 2 + 5 \operatorname{cosec} x - 10 = 0$$

$$2 \operatorname{cosec}^2 x + 5 \operatorname{cosec} x - 12 = 0$$

AG

A1

2

(ii) $(2 \operatorname{cosec} x - 3)(\operatorname{cosec} x + 4) = 0$

Attempt to solve

M1

$$\operatorname{cosec} x = \frac{3}{2} \text{ or } -4$$

Condone answers with no method shown

A1

$$\sin x = \frac{2}{3} \text{ or } -\frac{1}{4}$$

AG

A1

3

(b) $(\theta - 0.1) = 0.73, 2.41, -0.25, -2.89$ AWRT

*2 correct values, may be implied later
(41.8, 138.2, -165.5, -14.5)*

B1

$\theta = 0.83, 2.51, -0.15, -2.79$ AWRT

2 correct answers

B1

+ 2 correct answers and no extra within range

B1

3

[8]

Q7.

(a) $\cos 2x = 1 - 2\sin^2 x$

B1

1

(b) (i) $3\sin x - \cos 2x = 3\sin x - (1 - 2\sin^2 x)$

Candidate's $\cos 2x$ or $\sin^2 x$

M1

$= 3\sin x - 1 + 2\sin^2 x$

AG

A1

2

(ii) $2\sin^2 x + 3\sin x - 2 = 0$

Soluble quadratic form

M1

$(2\sin x - 1)(\sin x + 2) = 0$

Attempt to solve (allow one error in formula, allow sign errors)

M1

$\sin x = \frac{1}{2} \quad x = 30 \quad x = 150$

M1

$\sin^{-1}$ and two solutions ($0^\circ < x < 360^\circ$)

A0 if radians

A1

4

Allow misread for $2\sin^2 x + 3\sin x - 1 = 0$

Soluble quadratic form

(M1)

$$\sin x = \frac{-3 \pm \sqrt{17}}{4}$$

Use of formula (allow one error)

(M1)

$$x = 16.3^\circ, 163.7^\circ$$

Max 3/4

(A1)

(c) $\int \frac{1}{2}(1 - \cos 2x) = \frac{x}{2} - \frac{\sin 2x}{4} (+c)$

*M1 – solve integral, must have 2 terms
for $\sin^2 x$ from (a)*

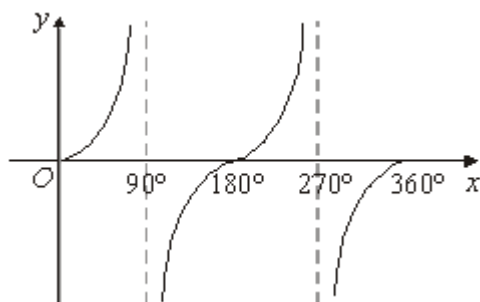
M1A1

2

[9]

Q8.

(a)



*Correct shape of branch from O {to 90°}
or correct shapes of branches from 90°- 360°*

M1

*Complete graph for $0^\circ \leq x \leq 360^\circ$
(Asymptotes not explicitly required but graphs
should show "tendency")*

A1

Correct scaling on x-axis $0^\circ \leq x \leq 360^\circ$

A1

3

(b) 61° ;

For 61°

B1

241°

For 241° and no 'extras' in the interval $0^\circ \leq x \leq 360^\circ$

B1

2

(c) (i) $\sin \theta = -\cos \theta \Rightarrow \frac{\sin \theta}{\cos \theta} = -1$

$\Rightarrow \tan \theta = -1$

AG; be convinced that the identity $\frac{\sin \theta}{\cos \theta} = \tan \theta$ is known and validly used

B1

1

(ii) $\Rightarrow \tan(x - 20^\circ) = -1$

M1

$x - 20^\circ = \tan^{-1}(-1)$

m1

$x - 20^\circ = 135^\circ, 315^\circ \dots$

$x = 155^\circ;$

A1

335°

Ft on $(180 + "155")$ and no 'extras' in the given interval.

A1ft

4

(d) Translation

'Translation'/'translate(d)'

B1

$$\begin{bmatrix} 20 \\ 0 \end{bmatrix}$$

Accept equivalent in words provided linked to 'translation/move/shift' (Note: B0B1 is possible)

B1

2

(e) $f(x) = \tan 4x$

For $\tan 4x$

B1

1

[13]

Q9.

(a) $\operatorname{cosec} x = 2$

$$\Rightarrow \sin x = \frac{1}{2}$$

30° scores M1 implied

M1

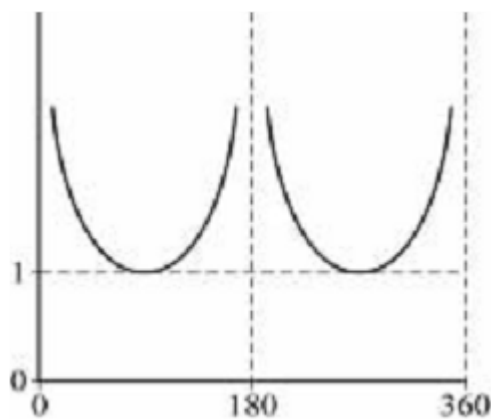
$= 30, 150$

and no extras in range

A1

2

(b)



all positive, 2 U shapes

M1

*minima consistent > 0 , **not** intersecting with each other or y-axis*

A1

2

[4]

Q10.

(a) $R = 5$

B1

$$\tan \alpha = \frac{3}{4} \text{ (OE)} \quad \alpha = 36.9^\circ \text{ (ISW 216.9)}$$

$$\text{SC1 } \tan \alpha = \frac{4}{3}, \alpha = 53.1^\circ$$

R, \alpha, PI in (b)

M1A1

3

(b) $\cos(x - \alpha) = \frac{2}{R}$

M1

$$x - \alpha = 66.4^\circ$$

A1

$$x = 103.3^\circ$$

A1F

$$x = 330.4^\circ$$

*accept 330.5°, -1 each extra
ft on acute \alpha*

A1F

4

(c) minimum value = -5
ft on R

B1F

$$\cos(x - 36.9) = -1$$

SC cos (x + 36.9) treat as miscopy

M1

$$x = 216.9^\circ$$

*216.9 or better accept graphics calculator
solution to this accuracy*

SC Find max:

max = 5 at (x + 36.9) + stated 1/3

Max 8/10 for work in radians

A1

3

[10]

Q11.

- (a) (i) Stretch **(I)** in y -direction **(II)**

Scale factor 2 **(III)**

>1 transformation is M0.

*M1 for **(I)** and either **(II)** or **(III)** or **(III)***

M1A1

2

- (ii) Reflection;

M1

in x -axis

'Reflection'/'reflect(ed)'

(or in y -axis or $y = 0$ or $x = 0$)

A1

2

- (iii) Translation;

'Translation'/'translate (d)'

B1

$$\begin{bmatrix} 30^\circ \\ 0 \end{bmatrix}$$

*Accept **full** equivalent in words provided linked to 'translation/move/shift' and **positive** x -direction
(Note: B0 B1 is possible)*

B1

2

- (b) $\{\theta - 30^\circ = \sin^{-1}(0.7) = 44.4\dots^\circ$

Inverse sine of 0.7 PI eg by sight of 44, 74 or better

M1

..... = $180^\circ - 44.4^\circ$

Valid method for 2nd angle

m1

$\theta = 74.4^\circ, 165.6^\circ$

Condone >1dp accuracy

A1

3

- (c) ... = $\cos^2 x + 2\cos x \sin x + \sin^2 x + \cos^2 x - 2\cos x \sin x + \sin^2 x$

Award for either bracket expanded correctly

M1

$$\dots = 2\cos^2 x + 2\sin^2 x$$

OE

A1

$$= 2(\cos^2 x + \sin^2 x) = 2 (1)$$

$\cos^2 x + \sin^2 x = 1$ stated or used.

M1

$$= 2$$

AG (be convinced)

A1

4

[13]

Q12.

(a) $2\operatorname{cosec}^2 x = 5(1 - \cot x)$

$$2 + 2 \cot^2 x = 5 - 5 \cot x$$

use of $\operatorname{cosec}^2 x = 1 + \cot^2 x$

M1

$$2 \cot^2 x + 5 \cot x - 3 = 0$$

AG

A1

2

(b) $(2 \cot x - 1)(\cot x + 3) = 0$

or $2 + 5t - 3t^2 = 0$ Or in $\tan x$

$$(2 - t)(1 + 3t) = 0$$

M1

$$\cot x = \frac{1}{2}, -3$$

$$\tan x = 2, -\frac{1}{3}$$

AG

A1

2

(c) $x = 1.1, -2.0$ AWRT

$x = -0.3, 2.8$ AWRT

Any 2 correct (In degrees): B0

Any 3 correct (In degrees): B1

4 correct (In degrees): B2

B1

B1

B1

[7]

Q13.

(a) $R = \sqrt{13}$ Or 3.6

B1

1

(b) $\frac{\sin \alpha}{\cos \alpha} = \tan \alpha = \frac{2}{3}$ $\alpha \approx 33.7$

Allow M1 for $\tan \alpha = \frac{-2}{3}$ or $\pm \frac{3}{2}$
 AG convincingly obtained

M1A1

2

(c) maximum value = $\sqrt{13}$

B1F

$\cos(\theta + 33.7) = 1$ ($\theta = -33.7$)

M1

$\theta = 326.3$

AWRT 326

A1

3

[6]

Q14.

- (a) Stretch (I) in x -direction (II) scale
 Need(I) and one of (II),(III)

M1

factor 2 (III)

M0 if more than one transformation

A1

2

(b) $\tan^{-1} 3 = 1.2(49....) (= \alpha)$
 $\tan^{-1} 3$ [PI by 71.(56..)°]

M1

$$\left\{ \frac{1}{2}x \right\} = \pi + \alpha ;$$

Correct quadrant; condone degrees or mix

m1

$$\frac{1}{2}x = 1.249...; 4.3906...$$

$$x = 2.498... = 2.50 \text{ to 3 sf}$$

A1

$$x = 8.781... = 8.78 \text{ to 3 sf}$$

A1

*Condone 2.5 otherwise deduct **max** of 1 mark throughout this question from A marks if 'correct' rads. but to 2sf or final answers in degrees. (143°, 503°)*

As usual, accept greater accuracy answers. Ignore extra values outside the given interval (0 to 12.6). If > 2 values inside interval lose an A mark for each one.

NB M1m0A1A0 is possible

SC after M0 for error $\tan x = 6$;

Either $x = 1.40(5), 4.54(7), 7.68(8), 10.8(3)$ or $x = 80.5^\circ, 260.5^\circ, 440.5^\circ, 620.5^\circ$ SC
 B1

(accept each rounded or truncated to 3 sf)

4

(c) $\cos \theta = 0$, $\sin \theta - 3 \cos \theta = 0$
Need both

M1

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ or } \tan \theta = 3$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} \text{ seen/used}$$

M1

$$\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2} = 1.57(07\dots)$$

Accept $\frac{\pi}{2}$

B1

$$\text{or } \theta = \frac{3\pi}{2} = 4.71(23\dots)$$

Accept $\frac{3\pi}{2}$

B1

$$\tan \theta = 3 \Rightarrow \theta = 1.249\dots; 4.3906\dots = 1.25, 4.39 \text{ to 3sf}$$

If not correct, ft on (b)

NB M0M1(B0B0)A1ft is possible

A1 ✓

$$90^\circ; 270^\circ; 71.5(6)^\circ; 251.5(6)^\circ$$

5

[11]

Q15.

(a) $\sec x = 5$
 $\cos x = 0.2$

M1

$$x = 1.37, 4.91 \text{ AWRT}$$

A1A1

3

(b) $\tan^2 x = 3 \sec x + 9$
 $\sec^2 x - 1 = 3 \sec x + 9$

for using $\sec^2 x = 1 + \tan^2 x$ OE

M1

$$\sec^2 x - 3 \sec x - 10 = 0$$

AG

A1

2

- (c) $(\sec x - 5)(\sec x + 2) = 0$
or use of formula (attempt)

M1

$$\sec x = 5, -2$$

A1

$$\cos x = 0.2, -0.5$$

$$x = 1.37, 4.91$$

any 2 correct or ft their 2 answers in (a)

B1F

$$2.09, 4.19$$

all 4 correct, no extras

A1

4

[9]

Q16.

(a) (i) $\sin 2x = 2 \sin x \cos x$

B1

1

(ii) $\cos 2x = 2 \cos^2 x - 1$

B1

1

(b) $\sin 2x - \tan x = 2 \sin x \cos x - \frac{\sin x}{\cos x}$

Use of their $\cos 2x$ or $\sin 2x$

M1

$$= \sin x \left(2 \cos x - \frac{1}{\cos x} \right)$$

Use of $\tan x = \frac{\sin x}{\cos x}$ *and the other double angle identity*

M1

$$= \sin x \left(\frac{2 \cos^2 x - 1}{\cos x} \right) = \tan x \cos 2x$$

AG convincingly obtained

A1

3

(c) $\tan x \cos 2x = 0 \quad x = 180$

Ignore $x = 0$, $x = 360^\circ$ & any others outside range

B1

$$\cos 2x = 0 \text{ or } \cos^2 x = \frac{1}{2} \left(\text{or } \sin^2 x = \frac{1}{2} \right)$$

M1

$$x = 45$$

A1

$$x = 135, 225, 315$$

CAO max 3/4 for answers in radians

B1

4

[9]

Q17.

(a) $A(0^\circ, 1)$

Condone radians

B1

$$B(45^\circ, 0)$$

Condone (0.785, 0) or better.

B1

$$C(270^\circ, -1)$$

B1 for 270; B1 for -1

B1, B1

4

(b) Stretch (I) in x -direction (II) with a scale factor $\frac{1}{2}$ (III)

More than one transformation is M0

*M1 for (I) **and** either (II) or (III)*

M1A1

2

- (c) $\cos^{-1}0.37 = "68.284..." (= \alpha)$
Cos⁻¹0.37 (PI eg by 68.3 or 1.19)

M1

$$x = \frac{\alpha}{2} = 34.1(42.)^\circ$$

Condone 34.2°, 34° or 0.596 rads

A1

$$x = 180 - \frac{\alpha}{2}$$

OE eg $2x = 360 - \alpha$

m1

$$x = 180 + \frac{\alpha}{2} \quad \text{and} \quad x = 180 + 180 - \frac{\alpha}{2}$$

*OE Need both (OE for $2x =$)
 with no extras (quadrants)
 within the given interval*

m1

$$2x = 38.284...; 291.175...; \\ 428.284...; 651.715...$$

$$x = (34.1^\circ;)$$

$$145.9^\circ; 214.1^\circ; 325.9^\circ$$

*Dep. on **all** three method marks. Must be in degrees*

A1

5

[11]

Q18.

(a) (i) $4 \frac{\sin \theta}{\cos \theta} \sin \theta = 15$

$$\Rightarrow 4 \sin^2 \theta = 15 \cos \theta$$

AG *Be convinced*

B1

1

(ii) $\sin^2 \theta + \cos^2 \theta = 1$

OE seen

M1

$$4(1 - \cos^2 \theta) = 15 \cos \theta$$

$$4 \cos^2 \theta + 15 \cos \theta - 4 = 0$$

AG *Be convinced*

A1

2

(b) (i) $(4c - 1)(c + 4) = 0$

Factorisation or formula or completion of square

M1

$$c = -4, c = \frac{1}{4}$$

Both values

A1

2

(ii) Since $-1 \leq \cos \theta \leq 1$, the only possible value for $\cos \theta$ is $\frac{1}{4}$

AG *convincingly explained*

(Condone strict inequalities)

ft provided candidates answers for c

are $\frac{1}{4}$ and a value

k such that $k > 1$ or $k < -1$

E1ft

1

(iii) $\theta = 75.5^\circ$

B1

$$\theta = 284.5^\circ$$

*Ft on $[360 - c's 75.5^\circ]$ as only other
solution in the given interval*

B1ft

2

(c) $\Rightarrow \cos 4x = \frac{1}{4}$

Links with previous parts. P1

M1

$$x = 19^\circ, 71^\circ$$

ft on (iii)/4....(only ft if 2 answers in given range).

A1ft

2

[10]

Q19.

(a) $\tan^2 x = \sec x + 11$

Or attempt to form quadratic in $\cos^2$

$$\sec^2 x - 1 = \sec x + 11$$

$$\tan^2 x = \sec^2 x - 1$$

M1

$$\sec^2 x - \sec x - 12 = 0$$

AG

A1

2

(b) $(\sec x - 4)(\sec x + 3) = 0$

attempt at solving quadratic

M1

$$\sec x = 4, -3$$

A1F

$$\therefore \cos x = \frac{1}{4}, -\frac{1}{3}$$

$$\text{AG; (A0 if no use of } \cos x = \frac{1}{\sec x} \text{)}$$

A1

3

(c) $x = 76^\circ, 284^\circ$

2 correct

B1

$$x = 109^\circ, 251^\circ \text{ (or better)}$$

other answers (-1 each extra in range)

B1,B1

If radians $x = 1.32, 4.97$
 $1.91, 4.37$
B1 any 2 correct
B1 other 2 correct

[8]

Q20.

(a) $R = \sqrt{5}$ (or $\sqrt{1+2^2}$ or 2.23 or 2.24)

B1

$$\frac{\sin \alpha}{\cos \alpha} = \frac{1}{2} \quad \alpha = 26.6^\circ$$

CAO
SC 63.4° 1/2

M1A1

(b) $\sin(x + 26.6) = \frac{1}{\sqrt{5}}$

M1

$$x = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right) - 26.6$$

B1 $x = 0^\circ$

m1

$x = 0^\circ$ $x = 126.8^\circ$ (126.9°, or 127 or 126° with working)
lose if extra solution in range
SC calculator trace: 126.9 full marks

B1

SC ft from 63.4°

A1

[7]